

Python Data Structures Cheat Sheet

LIST

- Ordered, mutable, allows duplicates.
- Methods: `append()`, `extend()`, `insert()`, `remove()`, `pop()`, `clear()`, `index()`, `count()`, `sort()`, `reverse()`, `copy()`

Example:

```
my_list = [1, 2, 3]
```

```
my_list.append(4)    # [1, 2, 3, 4]
```

```
my_list.extend([5, 6]) # [1, 2, 3, 4, 5, 6]
```

SET

- Unordered, mutable, no duplicates.
- Methods: `add()`, `update()`, `remove()`, `discard()`, `pop()`, `clear()`, `union()`, `intersection()`, `difference()`, `symmetric_difference()`

Example:

```
my_set = {1, 2, 3}
```

```
my_set.add(4)
```

```
my_set.update([5, 6])
```

DICTIONARY

- Key-value pairs, unordered (ordered since Python 3.7), mutable, keys must be unique.
- Methods: `get()`, `keys()`, `values()`, `items()`, `pop()`, `popitem()`, `update()`, `clear()`

Example:

```
my_dict = {'a': 1, 'b': 2}
```

```
my_dict['c'] = 3
```

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```
my_dict.update({'d': 4})
```

PRACTICE PROBLEMS

1. Create a list of numbers and write a function to return only the even numbers.
2. Write a function that counts how many times each item appears in a list.
3. Remove all duplicates from a list using a set.
4. Given a dictionary, print keys that have even values.
5. Create a dictionary that maps each word in a sentence to its length.